

Software Architecture & End-to-End Logistics Pipeline Manual

A Comprehensive Engineering Guide to Bueno OS: Railway Heavy Freight Corridor Logistics, Monotonic Data Engine, Siding Operations, Live Telemetry & Financial Reconciliation

Target Audience:	Full-Stack Software Engineers, System Architects, Operations Leads, Finance Officers
Architecture:	Next.js 14 (App Router) + TypeScript + Tailwind CSS + PHP/MySQL Authoritative REST API
Corridor Operations:	Nigerian Railway Corporation (NRC) Western Corridor — Standard Gauge (1,435mm) & Narrow Gauge (1,067mm)
Fleet Capacity:	46 Dedicated PXG Covered Hopper Wagons (60 MT / 1,200 Bags per wagon consist)
Core Tenet:	Pure Monotonic Database Synchronization — Zero Mock/Local-Storage Drift — Authoritative Finance Costing
Version / Date:	Production Release v4.2 · September 2026 · Git Commit HEAD

Executive Summary & Document Purpose

This architectural specification is designed for any newly onboarded software developer, engineering manager, or operations analyst joining Bueno Logistics. It provides an exhaustive, code-level and process-level exposition of the entire Bueno OS platform. You will learn every nitty-gritty detail: how freight deals are structured and negotiated with industrial clients (e.g. Dangote Cement, HBM/Lafarge, Flour Mills), how trains are assembled and loaded wagon-by-wagon at siding hubs using live weighbridge slips and tamper-proof bolt seals, how multi-gauge locomotives are tracked along the Nigerian Railway corridor via real GPS telemetry, how destination cargo officers audit discharges and discrepancies, how the empty locomotive return run cycles back to origin base, and how the corporate Finance desk exercises sole authority over trip pricing, invoices, and operating profit margins.

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1. Business Mission & Railway Logistics Context

Bueno Logistics is a premier enterprise railway haulage and intermodal freight operator in Nigeria. The company bridges heavy industrial manufacturing sidings with inland distribution centers and maritime port terminals. Rather than relying on vulnerable, congestive long-haul highway trucking, Bueno utilizes the massive volume efficiencies of freight trains operated over Nigerian Railway Corporation (NRC) track infrastructure.

1.1 The Dual-Gauge Railway Infrastructure

A critical domain truth that every developer must understand is Nigeria's **dual-gauge railway architecture**. The system explicitly routes freight across two separate, non-interchangeable track gauges:

Track Parameter	Standard Gauge Network (SGR)	Narrow Gauge Network (Cape Gauge)
Track Gauge	1,435 mm (4 ft 8.5 in)	1,067 mm (3 ft 6 in)
Primary Corridor	Lagos Mobolaji Johnson (MBJ) → Papalanto (PAPA) → Abeokuta (AB) → Moniya Container Terminal, Ibadan (MNY/MONI)	Apapa Port (APT) → Ewekoro Siding (EWK) → Dugbe Station, Ibadan (DGB) → Osogbo (OSB) → Ilorin (ILR) → Kano (KAN)
Axle Load Limit	25.0 Metric Tonnes / Axle	16.0 to 18.0 Metric Tonnes / Axle
Locomotive Power	Mainline Diesel-Electric (L2205, L2208, CDD33A1 series)	Legacy Narrow-Gauge Locomotives (2101 / 2200 series)
Handling Cargo	Export/Import TEU Containers, Bulk Clinker, Bagged Cement	Bagged Portland Cement, Gypsum Raw Ore, Limestone, Heavy Steel Coils

1.2 The Rolling Stock Consist: 46 PXG Covered Hoppers

Bueno Logistics operates a dedicated core fleet of **46 official PXG Covered Hopper Wagons** (registered under codes PXG 09001 through PXG 09046). A covered hopper wagon is designed specifically to transport weather-sensitive bulk and bagged industrial cargo (such as cement bags and grain) or heavy raw mineral bulk (gypsum, clinker, limestone).

Consist Capacity Math: Exactly **1 Wagon = 1,200 Bags (50kg each) = 60 Metric Tonnes (MT)**. A standard commercial train consist dispatched on the corridor is typically composed of **20 to 23 Wagons** (carrying **24,000 to 27,600 Bags / 1,200 to 1,380**

MT). Every calculation in the system (tonnage, axle load, demurrage, tariff) flows mathematically from these physical consist realities.

2. System Architecture & The Monotonic Sync Engine

The Bueno platform is engineered as a high-reliability, responsive web application that functions flawlessly both on desktop command screens at Executive HQ and on ruggedized mobile tablets used by Cargo Officers standing on dusty siding tracks in Papalanto or Ewekoro.

2.1 Technology Stack Decomposition

Component Layer	Technology Choice	Architectural Responsibility
Frontend Framework	Next.js 14.2 (React 18, TypeScript)	Modular desk portals, real-time reactive state, SSR/SSG export, strict type validation.
Styling & Icons	Tailwind CSS + Lucide React	Enterprise corporate white/brand-green design system, custom responsive data tables.
State Engine	StateEngine.ts (Monotonic Singleton)	Centralized business logic, consist math, dynamic fleet allocation, date categorizations.
API Layer	PHP 8.2 Micro-REST Endpoints	Stateless JSON endpoints on cPanel server (<code>/public/api/*.php</code>), atomic DB transactions.
Persistence Store	MySQL Relational Database	Exclusive source of truth for deals, trips, wagon logs, invoices, role permissions.
Telemetry & GPS	Leaflet.js + Haversine Engine	Live train coordinate interpolation, distance calculations, arrival geofence triggers.

2.2 The 'Pure Monotonic Database Sync' Principle

Critical Engineering Mandate: In early prototypes, client-side `localStorage` caching caused deleted test deals and mock trips to resurrect upon page reloads. This flaw was permanently eradicated with the **Pure Monotonic Sync Engine** implemented in `StateEngine.ts`:

- Database-Authoritative Overwrite:** When `StateEngine.syncRemote()` fetches trips or deals from `/api/trips.php` or `/api/deals.php`, the returned MySQL array completely overwrites local client state. No merge or reconciliation logic is permitted to re-insert phantom records.
- Zero Mock Clean Slate:** Default seed stores for deals and trips are strictly empty arrays (`[]`). Production launches start pristine.
- Atomic Purge Triggers:** Every PHP backend endpoint features an authenticated `{ action: 'PURGE_ALL' }` handler, allowing administrators to wipe test runs without manual SQL queries.
- Cache Invalidation Key:** An automatic version hash (`bueno_prod_release_v31_clean_slate`) clears residual browser cache whenever a new major build is deployed.

2.3 Codebase Organization

Developers should navigate the project following this directory structure:

```
apps/web/src/app/dashboard/page.tsx – Master gateway orchestrating role routing.
apps/web/src/components/portals/AdminPortal.tsx – 12-tab HQ Command Portal (Analytics, Deals, Siding, Fleet, Billing, RBAC).
apps/web/src/components/portals/CargoOfficerPortal.tsx – Field Officer Siding Console (Timed Loading, Feeder Trucks, Offload).
apps/web/src/components/portals/CustomerPortal.tsx – B2B Consignee Portal (Consignment Requisitions, Live GPS, Invoices).
apps/web/src/components/TerminalInformationView.tsx – 16-column Official Siding Ledger with Start/End/Duration CSV download.
apps/web/src/lib/services/StateEngine.ts – Central monotonic state and mathematical domain engine.
apps/web/public/api/*.php – Direct backend MySQL REST API endpoints.
```

3. Database Schema & API Micro-Endpoints

All data persistence is backed by MySQL on the cPanel production server. Below is the structural schema definition of the core tables:

Table Name	Key Columns & Types	Functional Role in Architecture
bueno_deals	id (VARCHAR 64 PK), deal_number, company, cargo_type, quantity, unit_of_measure, tariff_rate_per_ton, loading_station, destination, gauge, deal_type, total_planned_trips, dispatched_trips_count, status	Stores contracted B2B freight commitments signed between Bueno and consignors. Governs planned trip tranches.
bueno_trips	id (VARCHAR 64 PK), trip_id, deal_id, locomotive_id, driver_name, origin, destination, cargo_type, quantity, target_wagons_count, status, wagon_logs (LONGTEXT JSON), cur_lat, cur_lng, speed, progress_percent, is_return_leg, parent_trip_id, finance_cost, costing_status	The core operational entity. Encapsulates physical train movement, consist wagon tallies, live GPS coordinates, and return runs.
bueno_invoices	id (VARCHAR 64 PK), invoice_number, trip_id, deal_id, company_name, route, subtotal, damage_units, damage_deduction, total_amount, amount_paid, balance, status, issue_date, due_date	Commercial customer billing ledger. Automatically reflects damage debit deductions and tracks AR remittances.
bueno_trip_costs	id (VARCHAR 64 PK), trip_id, category, title, vendor, amount (DECIMAL), voucher_no, payment_status, created_at	Direct Corridor COGS: NRC track access fees, locomotive AGO fuel, shunting tariffs, siding lashing labor.
bueno_role_permissions	role (VARCHAR 32 PK), permissions (LONGTEXT JSON), updated_at	Enterprise RBAC storage. Holds the 22 unified capabilities array per system role. Survives all client reloads.
bueno_chat_messages	id (VARCHAR 64 PK), deal_id, sender_name, sender_role, message, timestamp	Bilateral contract negotiation channel between consignors and commercial administrators.

3.1 The wagonLogs Consist JSON Structure

Inside each row of bueno_trips, the column wagon_logs stores an audited JSON array representing every individual wagon coupled to the locomotive. A developer must maintain this exact schema contract:

```
{ "id": "wL_1789651586_402", "wagonId": "PXG 09029", "status": "LOADED", // LOADING | LOADED | EMPTY "unloadStatus": "UNLOADED", // PENDING_UNLOAD | UNLOADING | UNLOADED "startDate": "17 Sept 2026", "startTime": "13:38:12", // Stopwatch loading start "endDate": "17 Sept 2026", "endTime": "14:15:30", // Stopwatch loading completion "durationStr": "37 Minutes", // Calculated elapsed duration "qty": 1200, // Real bag tally or bulk MT "unitOfMeasure": "Bags", "sourceEnv": "Silo Bay 1 - Ewekoro", "sealNumber": "SEAL-NG-99214", // Tamper-proof bolt seal reference "truckRegNo": "KJA-482-XY, BDG-119-ZZ", // Feeder truck license plates "driverDetails": "Ibrahim Garba (08031112233)", "unloadStartTime": "16:20:00", // Stopwatch offload start at destination "unloadEndTime": "17:02:15", // Stopwatch offload stop "unloadDurationStr": "42 Minutes", "damageQty": 0, // Verified torn/damaged units "burstBags": 0 // Verified burst bags at siding }
```

4. Enterprise RBAC & Unified Capabilities Matrix

Bueno OS enforces a strict **Role-Based Access Control (RBAC)** model aligned with enterprise Spatie standards. In earlier versions, screen tab permissions and backend action permissions were disconnected. We completely overhauled this foundation into a **Unified 22-Capability Matrix** stored in MySQL table `bueno_role_permissions`.

Capability Domain	Capability Key / Action String	Access Scope & Operational Meaning
Domain 1: Screen & Navigation	terminal_info, deals, negotiations, wallet, moniya, fleet, tracking, consignments, billing, users, permissions, analytics	Controls visibility and mounting of top-level command tabs in the left navigation sidebar.
Domain 2: Commercial Deals	deals.create, deals.edit, deals.dispatch, deals.pricing	Creating freight deals, editing volume quotes, approving tranches, and locking base contract tariffs.
Domain 3: Corridor & Siding	ops.dispatch, ops.loading_log, ops.seal_break, fleet.manage	Dispatching trains, starting/stopping wagon loading, recording feeder trucks, seal verification, wagon repairs.
Domain 4: Finance & Treasury	finance.invoices_issue, finance.payments_record, finance.trip_costing, finance.cogs_manage	Sole pricing authority on trips, official corporate invoice generation, damage claim adjustments, recording COGS.
Domain 5: System Administration	admin.audit_view, admin.matrix_manage, admin.users_provision	Configuring RBAC matrix, provisioning client/officer accounts, purging test data.

4.1 Role Assignment & Persona Profiles

The system provisions 9 canonical user roles, each assigned a tailored subset of the capabilities matrix:

- **ADMIN (Folake Adeyemi)**: Master Administrator with universal 22/22 capabilities, user provisioning, database purge, and matrix editing.
- **CEO (Alhaji Bashir Umar)**: Executive Strategic Command with read/oversight authority over all analytics, live GPS, deals, and reports.
- **HEAD_OF_OPERATIONS (Engr. Mustapha Bello)**: Full corridor operational command, train consist assembly, dispatch authorization, siding monitoring.
- **HEAD_OF_FINANCE / ACCOUNTANT (Ibrahim Sanusi)**: Sole pricing authority for trip costing, customer invoices, debit notes, COGS vouchers.
- **CARGO_OFFICER (Ade Bello)**: Station Siding Lead at Ewekoro/Papalanto. Timed wagon loading deck, feeder truck audits, seal logging, offload audits.
- **CUSTOMER / CONSIGNEE (Dangote, Lafarge, Flour Mills)**: B2B Consignee portal. Freight requisition, bilateral deal negotiations, live GPS, invoices.

5. End-to-End Logistics Lifecycle: The 8-Phase Flow

To build or maintain features in Bueno OS, a developer must master the complete freight lifecycle. Every consignment moves strictly through these 8 sequential, interconnected phases across user desks:

Phase 1: B2B Client Requisition [Customer Portal]

The industrial client (e.g. Dangote Cement) logs into `/auth/client`, selects origin siding (e.g. Ewekoro EWK) and destination (Dugbe DGB), specifies cargo classification ('Bagged Cement 50kg'), and enters requested haulage tonnage (e.g. 24,000 Bags / 1,200 MT). Submitting creates an authenticated Consignment Note Requisition in MySQL (`bueno_client_requests`).

Phase 2: Commercial Negotiation & Deal Signing [Commercial Desk (Admin)]

Commercial Admin reviews the requisition, evaluates track gauge compatibility (Narrow vs Standard), and initiates bilateral chat via `/negotiations.php`. Parties agree on tariff rate per ton (e.g. ■12,500/MT). Admin clicks '+ Create Freight Deal', selecting Contract Type (`SINGLE_TRIP` vs `MONTHLY_CONTRACT`). This establishes an active contractual deal record in MySQL (`bueno_deals`).

Phase 3: Tranche Scheduling & Trip Creation [Operations Command]

When cargo is ready at the factory silo, Operations clicks 'Dispatch Deal Tranche'. The system generates a new trip in MySQL (`bueno_trips`) in `LOADING` status with empty `wagonLogs: []`, assigning an initial locomotive (e.g. `L2205`) and crew. This immediately routes the active loading job to the Siding Cargo Officer's dashboard.

Phase 4: Siding Operations — Timed Wagon Loading [Origin Cargo Officer]

The Cargo Officer at Ewekoro siding opens the active trip. The loading deck dynamically displays only available wagons from the 46 PXG fleet. Officer selects wagon `PXG 09029` and clicks 'Start Loading' (records `startTime: 13:38:12`, silo bay). Third-party feeder trucks hauling cement from the factory enter the siding. Officer logs truck plate (`KJA-482-XY`), driver name, and weighbridge bag count. Upon filling (1,200 bags = 60 MT), Officer clicks 'Stop Loading' (records `endTime: 14:15:30`, calculates `durationStr: 37 Minutes`, and logs bolt seal `SEAL-99214`). Wagon status updates from `LOADING` to `LOADED_INTACT`. Process repeats concurrently across consist.

Phase 5: Corridor Rail Dispatch & Live GPS [Operations / Executive]

With all 20 wagons loaded and sealed, Officer clicks 'Dispatch Train & Activate GPS'. Trip status transitions to `IN_TRANSIT`. Real-time coordinates (`curLat: 6.8974`, `curLng: 3.2141`) begin streaming at ~55 km/h along the corridor. Executives, Operations, and Consignees track locomotive position, bearing, passing stations, and ETA on the interactive Leaflet GPS Map.

Phase 6: Destination Arrival & Discharge Audit [Destination Cargo Officer]

Upon crossing the destination geofence (e.g. Dugbe Siding, Ibadan), status updates to `ARRIVED`. Destination Cargo Officer conducts physical seal verification before discharge. Officer starts offload stopwatch (`unloadStartTime`). As bags are discharged into customer warehouse or local trucks, officer records physical counts. If 12 bags burst due to transit friction, officer logs `burstBags: 12` with damage notes. System automatically creates an insurance discrepancy alert (`bueno_notifications`). Wagons update to `UNLOADED`.

Phase 7: The Empty Train Return Flow [Cargo Officer / Siding Lead]

Once all wagons are unloaded, Cargo Officer is presented with two explicit choices: (1) 'Complete Consignment & Keep in Yard' or (2) '■ Dispatch Empty Return Run (Back to Base)'. Selecting the return run finalizes the laden leg (`TRIP-001`) as `COMPLETED` and automatically launches `TRIP-001-RET` from Dugbe back to Ewekoro with status `RETURNING\_EMPTY`, quantity 0 MT, and empty wagons. Locomotive tracks on live GPS back to origin base. Upon docking at Ewekoro siding, all 20 wagons automatically become `AVAILABLE` in fleet inventory for the next deal tranche!

Phase 8: Finance Billing & Cost Reconciliation [Finance & Treasury Desk]

The tail end of the logistics cycle. Finance desk in 'Commercial Invoices & Ledger' reviews the completed trip. Finance verifies negotiated rate, inspects recorded damage count, and enters the official agreed trip price (e.g. ■15,000,000). Finance applies agreed burst-bag debit deduction (e.g. 12 bags × ■8,000 = -■96,000). System issues formal corporate invoice with VAT/WHT breakdown (`bueno\_invoices`) and aggregates operating COGS (NRC track toll, diesel). Net corridor gross margin is calculated and posted to corporate P&L.;

6. Dynamic Wagon Fleet Management

A critical fix implemented in the platform was the elimination of hardcoded wagon availability badges. In earlier code, the Wagon Fleet table hardcoded `<Badge text='AVAILABLE' color='green' />`, meaning all 46 wagons appeared available even when 20 were coupled to an active train.

6.1 How StateEngine.getDynamicWagonFleet() Works

The dynamic fleet calculation scans all registered wagons and cross-references them against active trips in real time:

- Active trips are filtered: `trips.filter(t => t.status !== 'COMPLETED' && t.status !== 'CANCELLED')`.
- Wagons coupled to an active trip inherit the exact operational state of the consist:
 - If trip is in `LOADING`: wagon displays `LOADED (TRIP-001)` or `LOADING`.
 - If trip is in `IN_TRANSIT`: wagon displays `IN_TRANSIT (TRIP-001)` in amber.
 - If trip is in `UNLOADING`: wagon displays `UNLOADING (TRIP-001)` in purple.
 - If trip is in `RETURNING_EMPTY`: wagon displays `RETURNING EMPTY (TRIP-001-RET)` in blue.
- Active Assignment Column explicitly reports: `TRIP-001 • EWK → DGB (Bagged Cement 50kg)`.
- Unassigned wagons display `AVAILABLE` in green with station: `Standing at Siding (Uncoupled)`.
- Fleet KPI Cards dynamically calculate: **26 Available | 20 In-Use / Coupled | 46 Total Registered**.

6.2 Consist Mathematical Constraints

The system enforces railway mechanical consist limits:

- Single Wagon Payload:** 1,200 Bags of 50kg Cement = 60.0 Metric Tonnes.
- Maximum Train Consist:** 23 Wagons = 27,600 Bags = 1,380 Metric Tonnes.
- Track Axle Load Safety:** Train assembly prevents dispatch exceeding NRC corridor bridge limits.

7. Terminal Information Ledger & Siding Audit Engine

The **Terminal Information Ledger** (`TerminalInformationView.tsx`) is the official siding audit document matching Nigerian Railway Corporation regulatory standards. Every single data point in this ledger is 100% dynamic, derived directly from wagon loading stopwatch audits.

Col #	Field Name	Audit Data Source & Computation Rule
1	WAGON NO.	Official PXG carriage registration (e.g. PXG 09029)
2	CONDITION	LOADED_INTACT, GOOD, DISCHARGED, or DEFECTIVE
3	REMARK / SILO BAY	Loading bay origin (e.g. 'Silo Bay 1 - Loading Siding') and seal remarks
4	DATE	Actual calendar date when wagon loading was conducted
5	START TIME	Exact timestamp when officer clicked 'Start Loading' (e.g. 13:38:12)
6	END TIME	Exact timestamp when officer clicked 'Stop Loading' (e.g. 14:15:30)
7	DURATION	Calculated elapsed time (e.g. 37 Minutes or 1h 12m)
8	TRAIN / TRIP NO.	Assigned trip identifier (e.g. TRIP-001)
9	ORIGIN / DEST	Loading siding station code → Destination discharge station code
10	CONTENT	Commodity classification (e.g. 'Bagged Cement 50kg')
11	TONNAGE (MT)	Exact tonnage: <code>qty * 0.05 MT</code> for cement bags (e.g. 60 MT)
12	QUANTITY (BAGS)	Audited bag count entered from factory weighbridge slip (e.g. 1,200 Bags)

13	FEEDER TRUCK(S)	Feeder truck license plate(s) transferring cargo into wagon (e.g. KJA-482-XY)
14	WAYBILL NO.	Unique NRC & Bueno consignment waybill reference
15	SEAL NO.	Tamper-proof bolt seal serial number verified at departure
16	DEMURRAGE	Siding dwell demurrage penalty (■15,000/day after free siding window)

7.1 Per-Trip Filter & Dedicated CSV Export

Users can filter the ledger by individual station siding or select **Filter Trip: TRIP-001**. Selecting a trip displays an Audit Focus Card summarizing locomotive ID, route, consist size, and total payload. Clicking **'Download Trip Ledger (CSV)'** dynamically compiles and downloads an official CSV audit spreadsheet named BUENO_TRIP-001_TERMINAL_LEDGER_[timestamp].csv containing all 16 audit columns.

8. Finance Desk Authority: Costing, Invoices & COGS

Architectural Rule: Finance is the Exclusive Authority on Trip Costing. In earlier builds, automated formulas multiplied total bag counts by per-ton rates, fabricating an erroneous ■300,000,000 figure. This was completely eradicated. The system strictly forbids automated revenue invention. Every trip cost must be explicitly audited and confirmed by the Finance Desk.

8.1 The Trip Costing & Pricing Authority Desk

Located under **Commercial Invoices & Ledger** → **Trip Costing & Pricing Authority**, Finance has a dedicated workflow:

1. **Audited Trip Roster:** Finance reviews all active and completed trips with real consist volumes and verified offload damage reports.
2. **Costing Status:** Unpriced trips display an amber badge: AWAITING PRICING (Costing Pending). No revenue is posted.
3. **'Update Trip Cost' Modal:** Finance opens the pricing modal to enter:
 - **Agreed Total Freight Cost (■):** Negotiated lump sum (e.g. ■15,000,000).
 - **Tariff Rate Per Ton (■/MT):** Contractual corridor rate (e.g. ■12,500/MT).
 - **Damage Deductions (■):** Agreed debit deductions for burst bags logged by destination cargo officers.
 - **Commercial Payment Terms:** Corporate credit terms (e.g. 'Post-discharge 30-day corporate settlement').
4. **Authoritative Persistence:** Calling `StateEngine.updateTripFinancePricing()` authoritatively updates MySQL, sets `costingStatus` = 'COSTED', and instantly generates/updates the client's official invoice.

8.2 Customer Invoice Math & Burst Bag Debit Deductions

Customer invoices in `bueno_invoices` mathematically calculate:

- **Gross Freight Subtotal:** e.g. ■15,000,000.
- **Transit Damage Deduction:** e.g. 12 burst bags × ■8,000/bag = -■96,000.
- **Net Freight Payable:** ■15,000,000 - ■96,000 = ■14,904,000.
- **Applicable WHT & VAT:** 5% WHT credit deduction and 7.5% VAT computation.

8.3 Direct Corridor Operating COGS

In `bueno_trip_costs`, Finance records direct operating expenses incurred for that train run:

- **NRC Track Access Toll:** Standard corridor track usage tariff.
- **Locomotive Fuel (AGO Diesel):** Fuel liters consumed based on distance and gross train weight.
- **Siding Shunting & Tarpaulin Lashing:** Labor and crane handling expenses.
- **Net Corridor Gross Margin:** Net Billed Revenue - Direct COGS = Corridor Operating Profit.

9. Operational Intelligence & Analytics Engine

Tab 0 (**Executive Reports & Analytics**) in the Admin Portal was overhauled from a money-focused screen into a true **Executive Operational Intelligence & Supply Chain Analytics Dashboard**. It provides executives and operations directors with actionable operational throughput metrics.

9.1 The Four Core Operational Intelligence Cards

Card Metric	Display Value (Live Calculation)	Operational Significance
Total Freight Hauled	1,200 MT (Math.round(totalReportMT))	Net cargo tonnage delivered across active consignments. Replaces fake revenue.
Consignment Volume	24,000 Bags across 20 Wagons	Total unit throughput transported in covered hopper rolling stock.
Average Loading Speed	35 Mins / Wagon	Turnaround efficiency at the origin siding loading chute/silo.
Cargo Delivery Integrity	100.0% Intact (0 Defects / Bursts)	Quality compliance: percentage of bags delivered undamaged to consignee.

9.2 Elimination of the 1 / 10 Trains Target Benchmark Bug

The Bug Explained: The previous benchmark calculation contained `const target = Math.max(item.actual, 10);`. Consequently, even when a user intentionally registered a Single Trip (1 train), the denominator defaulted to 10, producing '1 / 10 Trains (10% Target)'.

The Fix: Target is now queried from the deal's real contractual planned tranches (`deal.totalPlannedTrips`). For a single trip dispatch, target is strictly 1, displaying **1 / 1 Train (100% Target Met)** with **1,200 MT** completed!

9.3 Network Route Flow Distribution

The analytics engine dynamically computes active network traffic share: e.g. Ewekoro Siding (EWK) → Dugbe Station (DGB): 100% of Network Flow (1,200 MT) with an interactive animated visual progress bar.

10. Developer Setup, Conventions & Troubleshooting

10.1 Local Development Workflow

To run and develop the Bueno platform locally:

1. Clone repository: `git clone https://github.com/samuelmohel/Bueno.git`
2. Navigate to web workspace: `cd apps/web`
3. Install dependencies: `npm install`
4. Start development server: `npm run dev` (runs on `http://localhost:3000`)
5. Compile production build: `npm run build` (verifies TypeScript types and exports static HTML/JS to `apps/web/out/`).

10.2 Production cPanel Deployment Pipeline

The production application is deployed on cPanel with an Apache/PHP web server:

1. Build output lives in `apps/web/out/`.
2. All backend PHP scripts live in `apps/web/public/api/*.php` and are mirrored into `apps/web/out/api/*.php`.
3. When ready to deploy: commit all changes and push to GitHub (`git push origin main`).
4. In **cPanel Git Version Control**: click **Update from Remote** followed by **Deploy HEAD Commit**.
5. In client browser: perform a hard refresh (`Ctrl + Shift + R`) to invalidate static chunks.

10.3 Golden Engineering Rules (Never Violate)

- **RULE 1: Never Hardcode Fallback Mock Numbers.** Never write `|| 10` for trip targets, never hardcode `<Badge text='AVAILABLE' />`, never invent money figures.
- **RULE 2: Finance Sole Pricing Authority.** Revenue numbers must only be rendered if set by the Finance Desk in MySQL.
- **RULE 3: Respect the Rail Consist Math.** 1 Wagon = 1,200 Bags = 60 Metric Tonnes. Never multiply bags by per-ton rates without converting bags to tons.
- **RULE 4: Always Keep PHP API Parity.** Any change made to `public/api/*.php` must be copied into `out/api/*.php` before deployment.
- **RULE 5: Database-Authoritative Overwrite.** Never merge local storage with server state in a way that allows deleted items to resurrect.

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